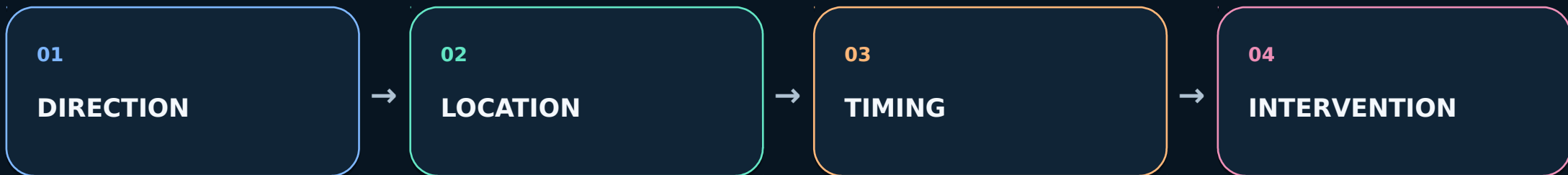


From behavioral association to a mechanistic claim

An evidence standard for deciding whether a sparse feature participates in value-guided choice.



CORE PRINCIPLE A feature earns a stronger name only when independent evidence agrees.

What would count as a value mechanism?

A directional correlation is the beginning of the analysis—not the conclusion.

SIGNED SUPPORT

$$D_f(x) = s(x)[\Delta_f, A(x) - \Delta_f, B(x)]$$

$$D_f(x) > 0$$

01

DIRECTION

The signed contrast follows the value-side action across option order.



02

LOCATION

Activity tracks the relevant option span rather than prompt scaffolding.



03

TIMING

Evidence appears before the answer, where it could influence the decision.



04

INTERVENTION

Ablation or patching changes the choice under matched controls.

CLAIM RULE Call it a candidate until direction, location, timing, and intervention converge.

One activation gap, three competing stories

A ranked feature is ambiguous until competing explanations are tested.

THE TEMPTING INTERPRETATION

“This feature represents the value.”

Observed: activation is larger when the model chooses the value-consistent option.

**Useful evidence—
but not yet identity or
causality.**

01

PROMPT SCAFFOLD

The feature tracks answer format, option order, or repeated instructions.

02

CONTENT MATCH

The feature recognizes words in one option without participating in choice.

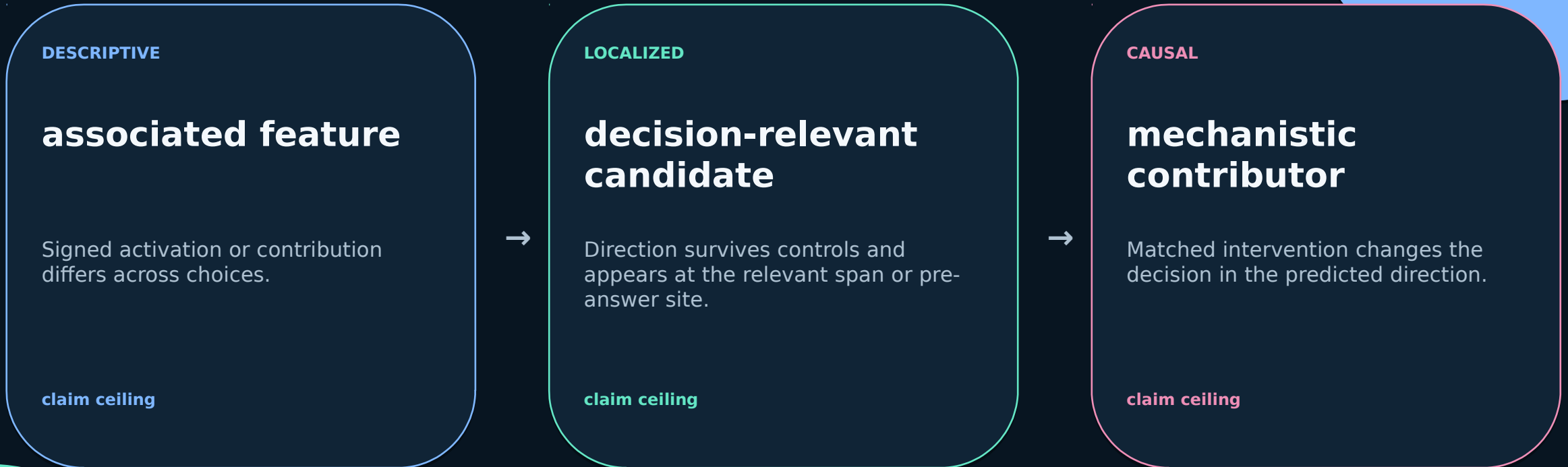
03

DECISION CONSEQUENCE

The feature activates after the answer direction is already determined.

Match the claim to the evidence

The vocabulary should change as the evidence stack becomes more demanding.



Mixed evidence is a result: preserve disagreement instead of averaging it into certainty.